

Sharp C^r Spectral Disks for the Rvachev–Thue–Morse Transfer Operator

Universal dyadic Markov disks, endpoint-jet ladders,
finite resonances, and optimal operator asymptotics

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Abstract

The Fourier-decay branch of the PROVEIT repository asks for the spectrum of its Rvachev–Thue–Morse transfer operator on finite-smoothness spaces. The analytic Hardy-space spectrum had collapsed to three points, and a later Sobolev manuscript found a regularity-dependent essential disk, but the periodic Hölder and C^r cases were left open. This article resolves the integer C^r problem exactly, both on the circle and on the interval, and proves a more general structural theorem for smooth dyadic Markov transfer operators.

For a smooth weight $0 \leq \omega \leq 1$ satisfying $\omega(x+1) = 1 - \omega(x)$, define

$$(\mathcal{P}_\omega f)(x) = \omega(x)f(x/2) + (1 - \omega(x))f((x+1)/2).$$

On $C^r(\mathbb{T})$, with $r \in \mathbb{N} \cup \{0\}$, its Fredholm essential spectrum is *always* the exact disk $\overline{D}(0, 2^{-r})$. Every point of the open disk is an eigenvalue of infinite geometric multiplicity. The proof is constructive: the doubling composition operator is a right inverse, an explicit infinite-dimensional branch-cancellation space supplies defect vectors, and differentiation identifies the Calkin image at order r with 2^{-r} times the continuous-space image. Exterior generalized eigenvectors automatically bootstrap to C^∞ .

On $C^r([0, 1])$, unmatched endpoint jets contribute the explicit finite ladder $\{\omega(0)2^{-j} : 0 \leq j \leq r\}$ and no other spectral values. For the Rvachev sine mask $\omega(x) = \sin^2(\pi x/2)$ this ladder collapses to zero. Writing $\mathcal{L} = \mathcal{P}_\omega/2$, we obtain the complete formula

$$\sigma_{\text{ess}}(\mathcal{L}; C^r) = \overline{D}(0, 2^{-r-1}), \quad \sigma(\mathcal{L}; C^r) = \overline{D}(0, 2^{-r-1}) \cup \{1/2, -1/4\}$$

on both the circle and the interval. The leading resonance is simple; once $r \geq 2$, the resonance $-1/4$ is isolated, semisimple, and two-dimensional. For $r \geq 2$ this gives a sharp two-resonance operator expansion, while a Calkin-algebra lower bound proves that no compact correction can improve the residual exponential rate 2^{-r-1} . The results are conventional mathematical proofs in an unrefereed manuscript; no new Lean declaration or claim of global publication priority is made.

Keywords. Rvachev up-function; Fabius function; transfer operator; essential spectrum; Fredholm theory; C^r regularity; endpoint jets; Thue–Morse; spectral resonance.

Claim boundary. The paper proves exact theorems relative to the operator and function spaces stated here. It resolves the integer C^r portion of a question explicitly left open in the inspected PROVEIT branch. It does not claim that every specialization is globally new, does not settle noninteger Hölder spaces, and does not constitute a machine-checked formalization. The repository pin, source blobs, external analytic inputs, and finite verification scope are recorded in [sections B and C](#).

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1 The repository question and the result

1.1 A delimited open target

The PROVEIT Fabius-function research branch studies the centered Rvachev up-function, its infinite sinc product, Thue–Morse identities, exact moments, and several transfer operators. The manuscript *Spectral Collapse and Alternating RMS Asymptotics for the Rvachev Up-Function* proves a finite analytic spectrum and then asks for the complete spectrum or essential spectral radius on C^r , Hölder, Sobolev, and endpoint-weighted spaces. Its warning is mathematically important: an analytic Fredholm determinant cannot simply be transferred to a weaker norm. A subsequent manuscript answers the periodic Sobolev part and the integer-order interval Sobolev part, but explicitly leaves Hölder and C^r spaces untreated [1, 2].

This article answers the integer C^r problem. The answer is not the analytic three-point set. At every finite differentiability order there is a full essential disk. Its radius is particularly simple in the supremum-jet geometry of C^r : one factor $1/2$ comes from the repository normalization, and one factor 2^{-r} comes from the contraction of the highest derivative. The two algebraic resonances survive outside that disk exactly when their moduli exceed this threshold.

The proof also isolates a general mechanism. It applies to every smooth nonnegative dyadic partition of unity, not only to the sine mask. The essential disk comes from three pieces that fit exactly:

1. a right inverse generated by the doubling map;
2. an infinite-dimensional branch-cancellation kernel;
3. a derivative quotient that conjugates the Calkin image at order r to 2^{-r} times the continuous-space image.

The sine mask is then special only in the finite resonance calculation and in its vanishing endpoint value, which kills the interval jet ladder.

1.2 Repository normalization

The operator appearing in the Rvachev RMS analysis is

$$(\mathcal{L}f)(x) = \frac{1}{2} \left[\sin^2\left(\frac{\pi x}{2}\right) f(x/2) + \cos^2\left(\frac{\pi x}{2}\right) f((x+1)/2) \right]. \quad (1)$$

It is convenient to remove the outer factor and write

$$\mathcal{P} = 2\mathcal{L}. \quad (2)$$

Thus $\mathcal{P}\mathbf{1} = \mathbf{1}$ and $\mathcal{P}$ is a positive Markov operator in the supremum norm. All structural proofs are first given for $\mathcal{P}$ and then scaled back to $\mathcal{L}$.

We regard $C^r(\mathbb{T})$ as a complex Banach space with norm

$$\|f\|_{C^r} = \sum_{j=0}^r \|f^{(j)}\|_{\infty}, \quad r \in \mathbb{N} \cup \{0\}. \quad (3)$$

Equivalent standard C^r norms give the same spectrum. On the interval we use the analogous norm on $C^r([0, 1])$. Essential spectrum always means the Fredholm essential spectrum,

$$\sigma_{\text{ess}}(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not Fredholm}\}. \quad (4)$$

1.3 Main theorem in the Rvachev case

Theorem 1.1 (Complete C^r spectrum of the Rvachev operator). *Let $r \in \mathbb{N} \cup \{0\}$ and let $\mathcal{L}$ be (1). On both $C^r(\mathbb{T})$ and $C^r([0, 1])$,*

$$\sigma_{\text{ess}}(\mathcal{L}) = \overline{D}(0, 2^{-r-1}), \quad (5)$$

$$\sigma(\mathcal{L}) = \overline{D}(0, 2^{-r-1}) \cup \{1/2, -1/4\}. \quad (6)$$

Every λ with $|\lambda| < 2^{-r-1}$ is an eigenvalue of infinite geometric multiplicity.

The following exterior multiplicity statements hold.

- (a) *If $r \geq 1$, then $1/2$ is isolated and simple, with eigenspace $\text{span}\{\mathbf{1}\}$.*
- (b) *If $r \geq 2$, then $-1/4$ is isolated and semisimple with algebraic and geometric multiplicity two. Its eigenspace is*

$$\text{span} \left\{ \sin(2\pi x), \cos(2\pi x) + \frac{1}{3} \right\}. \quad (7)$$

At $r = 1$, the value $-1/4$ lies on the essential boundary; at $r = 0$ it lies strictly inside the essential disk. The displayed smooth eigenfunctions remain valid there, but no finite isolated multiplicity is asserted.

The threshold picture is therefore exact.

r	Essential radius	Status of $1/2$	Status of $-1/4$
0	$1/2$	essential boundary	interior of essential disk
1	$1/4$	isolated, simple	essential boundary
$r \geq 2$	2^{-r-1}	isolated, simple	isolated, semisimple, rank 2

The finite analytic spectrum from the predecessor is not contradicted. It lives on a much smaller function space. The present disk quantifies exactly what finite smoothness restores.

1.4 Sharp operator asymptotics

The spectral set immediately gives an operator-level consequence that is stronger than a mere upper bound.

Corollary 1.2 (Two-resonance expansion and optimal residual rate). *On $C^r(\mathbb{T})$ or $C^r([0, 1])$ with $r \geq 2$, let Π_+ and Π_- be the Riesz projections of $\mathcal{L}$ at $1/2$ and $-1/4$. Then*

$$\mathcal{L}^n = 2^{-n}\Pi_+ + (-1/4)^n\Pi_- + R_n, \quad (8)$$

where $\text{rank } \Pi_+ = 1$, $\text{rank } \Pi_- = 2$, and for every $\varepsilon > 0$ there is $C_{r,\varepsilon}$ such that

$$\|R_n\|_{C^r \rightarrow C^r} \leq C_{r,\varepsilon}(2^{-r-1} + \varepsilon)^n. \quad (9)$$

The exponential threshold cannot be improved by any compact correction: for every compact $K : C^r \rightarrow C^r$ and every $n \geq 1$,

$$\|\mathcal{L}^n - K\|_{C^r \rightarrow C^r} \geq 2^{-(r+1)n}. \quad (10)$$

In particular, (10) applies to the finite-rank resonance sum in (8).

For $r = 1$ there is still a leading rank-one decomposition, but the alternating value $-1/4$ belongs to the essential boundary. Thus the scalar alternating RMS law in the repository and the operator-norm geometry genuinely separate at the critical differentiability order.

2 A universal dyadic Markov family

2.1 Weights and circle compatibility

Let $\omega \in C^\infty(\mathbb{R})$ satisfy

$$0 \leq \omega(x) \leq 1, \quad \omega(x+1) = 1 - \omega(x). \quad (11)$$

Define, initially for a 1-periodic function f ,

$$(\mathcal{P}_\omega f)(x) = \omega(x)f(x/2) + (1 - \omega(x))f((x+1)/2). \quad (12)$$

Although the two summands are not separately periodic, their sum is: using (11) and periodicity of f gives $\mathcal{P}_\omega f(x+1) = \mathcal{P}_\omega f(x)$. Thus (12) defines a bounded operator on every $C^r(\mathbb{T})$.

The Rvachev mask is

$$\omega_R(x) = \sin^2(\pi x/2), \quad 1 - \omega_R(x) = \cos^2(\pi x/2). \quad (13)$$

It satisfies $\omega_R(0) = 0$, a fact that will eliminate the interval endpoint ladder.

2.2 The universal theorem

Theorem 2.1 (Universal C^r essential disk). *Assume (11). For every integer $r \geq 0$,*

$$\sigma_{\text{ess}}(\mathcal{P}_\omega; C^r(\mathbb{T})) = \overline{D}(0, 2^{-r}). \quad (14)$$

Every λ with $|\lambda| < 2^{-r}$ is an eigenvalue of $\mathcal{P}_\omega$ on $C^r(\mathbb{T})$ with infinite geometric multiplicity. Furthermore:

- (i) *the powers of $\mathcal{P}_\omega$ are uniformly bounded on each $C^m(\mathbb{T})$;*
- (ii) *every spectral value with $|\lambda| > 2^{-r}$ is an isolated eigenvalue of finite algebraic multiplicity;*
- (iii) *every generalized eigenvector belonging to such a value lies in $C^\infty(\mathbb{T})$.*

The exact disk, rather than merely its radius, is the key point. The proof is split into four transparent steps: a right inverse, an explicit defect space, a continuous-space disk, and an exact Calkin conjugacy under differentiation.

3 Right inverse, defect vectors, and the continuous disk

3.1 The doubling right inverse

Let $T_2x = 2x \pmod{1}$ and define

$$(\mathcal{U}f)(x) = f(T_2x). \quad (15)$$

Then the partition-of-unity identity in (12) gives an exact one-sided inverse.

Lemma 3.1 (Right inverse). *On every periodic function space on which the formulas make sense,*

$$\mathcal{P}_\omega \mathcal{U} = \text{I}. \quad (16)$$

Moreover, for $0 \leq j \leq r$ and $n \geq 0$,

$$(\mathcal{U}^n f)^{(j)}(x) = 2^{nj} f^{(j)}(T_2^n x). \quad (17)$$

Proof. Both inverse branches are mapped by T_2 back to x modulo one. Hence

$$(\mathcal{P}_\omega \mathcal{U}f)(x) = \omega(x)f(x) + (1 - \omega(x))f(x) = f(x).$$

The derivative identity follows by repeated application of the chain rule. $\square$

3.2 An explicit infinite-dimensional kernel

Continuity and (11) imply that ω takes a value strictly between zero and one. Choose a nonempty open interval $I \Subset (0, 1)$ on which both branch weights are positive. For $g \in C_c^\infty(I)$, define $E_g \in C^\infty(\mathbb{T})$ by

$$E_g(y) = \begin{cases} (1 - \omega(2y))g(2y), & y \in I/2, \\ -\omega(2y - 1)g(2y - 1), & y \in (I + 1)/2, \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

The two support intervals are disjoint and compactly contained in the two half-circles, so the extension by zero is smooth.

Lemma 3.2 (Branch-cancellation defect). *For every $g \in C_c^\infty(I)$,*

$$\mathcal{P}_\omega E_g = 0. \quad (19)$$

The map $g \mapsto E_g$ is injective. Consequently $\ker \mathcal{P}_\omega \cap C^\infty(\mathbb{T})$ is infinite-dimensional.

Proof. For $x \in I$, the two branch values are

$$E_g(x/2) = (1 - \omega(x))g(x), \quad E_g((x + 1)/2) = -\omega(x)g(x),$$

so their weighted sum vanishes. Outside I , both values vanish. Positivity of $1 - \omega$ on I makes the first branch recover g , proving injectivity. $\square$

3.3 A disk of infinite-multiplicity eigenvalues

For $e \in \ker \mathcal{P}_\omega \cap C^\infty(\mathbb{T})$ and a complex number satisfying $|\lambda|2^r < 1$, consider

$$F_{\lambda,e} = \sum_{n=0}^{\infty} \lambda^n \mathcal{U}^n e. \quad (20)$$

By (17), the series converges absolutely in C^r .

Proposition 3.3 (Defect eigenseries). *If $|\lambda| < 2^{-r}$, then*

$$\mathcal{P}_\omega F_{\lambda,e} = \lambda F_{\lambda,e}. \quad (21)$$

For fixed λ , the map $e \mapsto F_{\lambda,e}$ is injective. Therefore every point of $D(0, 2^{-r})$ is an eigenvalue of infinite geometric multiplicity on $C^r(\mathbb{T})$.

Proof. Using $\mathcal{P}_\omega e = 0$ and (16),

$$\mathcal{P}_\omega F_{\lambda,e} = \sum_{n=1}^{\infty} \lambda^n \mathcal{U}^{n-1} e = \lambda F_{\lambda,e}.$$

Also $(I - \lambda \mathcal{U})F_{\lambda,e} = e$, so the map is injective. $\square$

At $r = 0$, positivity and the partition of unity give

$$\|\mathcal{P}_\omega f\|_\infty \leq \|f\|_\infty. \quad (22)$$

The preceding eigenseries now determines the entire continuous-space spectrum and essential spectrum.

Corollary 3.4 (Exact continuous-space disk). *On $C(\mathbb{T})$,*

$$\sigma(\mathcal{P}_\omega) = \sigma_{\text{ess}}(\mathcal{P}_\omega) = \overline{D}(0, 1). \quad (23)$$

Proof. The norm bound puts the spectrum in $\overline{D}(0, 1)$. Every point of the open unit disk has infinite-dimensional kernel for $\mathcal{P}_\omega - \lambda I$, so it belongs to the Fredholm essential spectrum. The non-Fredholm set is closed, therefore it contains the closed disk. $\square$

4 Differentiation and the exact Calkin image

4.1 The leading derivative identity

For $m \geq 1$, Leibniz' rule gives

$$D^m \mathcal{P}_\omega f = 2^{-m} \mathcal{P}_\omega (D^m f) + \mathcal{R}_m f, \quad (24)$$

where

$$\mathcal{R}_m f = \sum_{k=1}^m \binom{m}{k} 2^{-(m-k)} \omega^{(k)}(x) \left[f^{(m-k)}(x/2) - f^{(m-k)}((x+1)/2) \right]. \quad (25)$$

The displayed expression is periodic, even though its individual branch terms need not be.

Lemma 4.1 (Compact lower-jet remainder). *For $m \geq 1$, $\mathcal{R}_m : C^m(\mathbb{T}) \rightarrow C(\mathbb{T})$ is compact and annihilates constants.*

Proof. Every term in (25) uses derivatives of order at most $m-1$. Thus $\mathcal{R}_m$ factors through the compact embedding $C^m(\mathbb{T}) \hookrightarrow C^{m-1}(\mathbb{T})$, followed by a bounded branch operator. It annihilates constants because the two branch values then agree. $\square$

4.2 Removing constants

Let

$$Y = \left\{ g \in C(\mathbb{T}) : \int_0^1 g(x) dx = 0 \right\}, \quad (\mathcal{J}g)(x) = g(x) - \int_0^1 g(t) dt. \quad (26)$$

The map D^r induces a Banach-space isomorphism

$$\dot{D}^r : C^r(\mathbb{T}) / \text{span}\{\mathbf{1}\} \longrightarrow Y. \quad (27)$$

Indeed, a continuous zero-mean function can be integrated periodically r times, with integration constants chosen at each stage, and the periodic Poincare inequality gives boundedness of the inverse.

Define the compressed continuous operator

$$Q = \mathcal{J} \mathcal{P}_\omega|_Y : Y \rightarrow Y. \quad (28)$$

With respect to $C(\mathbb{T}) = \text{span}\{\mathbf{1}\} \oplus Y$, the continuous operator has an upper triangular block form

$$\mathcal{P}_\omega = \begin{pmatrix} 1 & \varphi \\ 0 & Q \end{pmatrix} \quad (29)$$

for a bounded functional φ . Finite-dimensional extensions do not change Fredholm essential spectrum, so [theorem 3.4](#) implies

$$\sigma_{\text{ess}}(Q) = \overline{D}(0, 1). \quad (30)$$

Proposition 4.2 (Exact Calkin conjugacy). *Let $\dot{\mathcal{P}}_{\omega,r}$ be the operator induced by $\mathcal{P}_\omega$ on $C^r(\mathbb{T}) / \text{span}\{\mathbf{1}\}$. Under the isomorphism (27),*

$$\dot{D}^r \dot{\mathcal{P}}_{\omega,r} (\dot{D}^r)^{-1} = 2^{-r} Q + K_r, \quad K_r \text{ compact on } Y. \quad (31)$$

Proof. The left-hand side applied to the class of f is represented by $D^r \mathcal{P}_\omega f$. Apply $\mathcal{J}$ to (24); this does not change the left side because a periodic derivative has zero mean. The leading term becomes $2^{-r} \mathcal{J} \mathcal{P}_\omega D^r f = 2^{-r} Q(D^r f)$, while $\mathcal{J} \mathcal{R}_r f$ is compact by [theorem 4.1](#). It descends to the quotient because $\mathcal{R}_r$ annihilates constants. $\square$

Proof of the exact disk in theorem 2.1. The one-dimensional invariant constant space does not change the Fredholm essential spectrum. Compact perturbations and similarity preserve it. Therefore (31) and (30) give

$$\sigma_{\text{ess}}(\mathcal{P}_\omega; C^r(\mathbb{T})) = 2^{-r} \sigma_{\text{ess}}(Q) = \overline{D}(0, 2^{-r}).$$

The infinite-multiplicity point spectrum in the open disk was proved in theorem 3.3. $\square$

This argument also yields the sharp compact-approximation obstruction.

Corollary 4.3 (Calkin lower bound). *For every $r, n \geq 0$ with $n \geq 1$ and every compact $K : C^r(\mathbb{T}) \rightarrow C^r(\mathbb{T})$,*

$$\|\mathcal{P}_\omega^n - K\|_{C^r \rightarrow C^r} \geq 2^{-rn}. \quad (32)$$

Proof. The Calkin image of $\mathcal{P}_\omega$ has spectrum $\overline{D}(0, 2^{-r})$. By spectral mapping, the spectrum of its n th power is $\overline{D}(0, 2^{-rn})$. The quotient norm dominates spectral radius, and the quotient norm of $\mathcal{P}_\omega^n$ is the infimum of $\|\mathcal{P}_\omega^n - K\|$ over compact K . $\square$

5 Exterior eigenvectors are automatically smooth

The Calkin theorem determines all non-Fredholm spectral values. To classify the remaining spectrum for a particular mask, one needs to know that exterior eigenvectors cannot hide at finite regularity. This follows from a simple bootstrap once power boundedness is established.

5.1 Uniform power bounds

Lemma 5.1 (Power boundedness on every integer C^m). *For each integer $m \geq 0$ there is a constant B_m such that*

$$\|\mathcal{P}_\omega^n f\|_{C^m} \leq B_m \|f\|_{C^m} \quad (n \geq 0). \quad (33)$$

Consequently the spectral radius of $\mathcal{P}_\omega$ on $C^m(\mathbb{T})$ is one.

Proof. The case $m = 0$ is (22). Suppose the assertion is known through order $m - 1$. Applying (24) to $\mathcal{P}_\omega^{n-1} f$ gives

$$\|D^m \mathcal{P}_\omega^n f\|_\infty \leq 2^{-m} \|D^m \mathcal{P}_\omega^{n-1} f\|_\infty + C_m \|\mathcal{P}_\omega^{n-1} f\|_{C^{m-1}}.$$

The induction hypothesis bounds the second term uniformly. Summing the geometric recurrence bounds the highest derivative; lower derivatives are already controlled. The eigenvalue one, carried by constants, shows that the spectral radius is exactly one. $\square$

5.2 Resolvent bootstrap

Proposition 5.2 (Smoothness of exterior generalized eigenspaces). *Fix $r \geq 1$ and $\lambda \in \mathbb{C}$ with $|\lambda| > 2^{-r}$. If*

$$(\mathcal{P}_\omega - \lambda I)^q f = 0 \quad (34)$$

for some $q \geq 1$ and $f \in C^r(\mathbb{T})$, then $f \in C^\infty(\mathbb{T})$.

Proof. First take $q = 1$. Put $u = D^r f$. From (24),

$$(\lambda I - 2^{-r} \mathcal{P}_\omega) u = \mathcal{R}_r f. \quad (35)$$

The right side lies in C^1 , because it only involves derivatives of f of order at most $r - 1$. By [theorem 5.1](#), the spectrum of $\mathcal{P}_\omega$ on both C and C^1 lies in the closed unit disk. Since $|\lambda| > 2^{-r}$, the operator on the left of (35) is invertible on both spaces. Its C^1 solution and its C solution agree by uniqueness, so $u \in C^1$ and $f \in C^{r+1}$.

Repeat at orders $r + 1, r + 2, \dots$. The strict inequality persists because $2^{-m} \leq 2^{-r}$ for $m \geq r$.

For a generalized eigenvector, induct on q . Set $g = (\mathcal{P}_\omega - \lambda I)f$. By induction, g is smooth. The same argument applied to the inhomogeneous equation $\mathcal{P}_\omega f - \lambda f = g$ has the additional smooth term $D^m g$ on the right and again raises regularity one derivative at a time. $\square$

5.3 Why exterior spectral values are eigenvalues

The complement of the disk (14) is connected and contains infinity. On that component, the Fredholm index of $\mathcal{P}_\omega - \lambda I$ is zero. Hence every spectral point outside the disk is an isolated eigenvalue of finite algebraic multiplicity. Together with [theorem 5.2](#), this completes the general assertions of [theorem 2.1](#).

6 Fourier collapse for the sine mask

We now specialize to (13). The finite Fourier support of the mask forces every exterior generalized eigenvector into a three-dimensional band.

6.1 Exact Fourier rule

Write $e_k(x) = e^{2\pi i k x}$. Since

$$\sin^2(\pi x) = \frac{1}{2} - \frac{1}{4}e_1(x) - \frac{1}{4}e_{-1}(x), \quad (36)$$

the standard roots-of-unity filter gives

$$\widehat{\mathcal{P}f}(k) = \widehat{f}(2k) - \frac{1}{2}\widehat{f}(2k-1) - \frac{1}{2}\widehat{f}(2k+1). \quad (37)$$

The absolute sum of the three coefficients is two.

After n iterations, every input frequency contributing to output frequency k has the form

$$q = 2^n k + \delta, \quad |\delta| \leq 2^n - 1, \quad (38)$$

and the total absolute coefficient sum is at most 2^n . Therefore, if $|k| \geq 2$, then every descendant satisfies

$$|q| \geq 2^n + 1. \quad (39)$$

Proposition 6.1 (Finite Fourier band for exterior chains). *Let $r \geq 1$, $|\lambda| > 2^{-r}$, and $(\mathcal{P} - \lambda I)^q f = 0$ with $f \in C^r(\mathbb{T})$. Then*

$$f \in F_1 := \text{span}\{e_{-1}, e_0, e_1\}. \quad (40)$$

Proof. By [theorem 5.2](#), f is smooth. First suppose $q = 1$ and write $a_k = \widehat{f}(k)$. From $\mathcal{P}^n f = \lambda^n f$, (39), and rapid Fourier decay, for every N and $|k| \geq 2$,

$$|\lambda|^n |a_k| \leq 2^n \sup_{|q| \geq 2^n + 1} |a_q| \leq C_N 2^{-n(N-1)}.$$

Choose N with $2^{1-N} < |\lambda|$ and let $n \rightarrow \infty$; then $a_k = 0$.

For a generalized chain, let $g = (\mathcal{P} - \lambda I)f$. By induction on the chain length, $g \in F_1$. The band F_1 is invariant. Therefore the high-frequency coefficients of $\mathcal{P}^n f$ still equal $\lambda^n a_k$ plus terms obtained from iterates of g , all of which vanish at $|k| \geq 2$. The same estimate gives $a_k = 0$. $\square$

6.2 The exact three-mode matrix

Using (37),

$$\mathcal{P}e_{-1} = -\frac{1}{2}e_{-1} - \frac{1}{2}e_0, \quad (41)$$

$$\mathcal{P}e_0 = e_0, \quad (42)$$

$$\mathcal{P}e_1 = -\frac{1}{2}e_0 - \frac{1}{2}e_1. \quad (43)$$

Thus, in the ordered basis (e_{-1}, e_0, e_1) ,

$$A = \begin{pmatrix} -1/2 & 0 & 0 \\ -1/2 & 1 & -1/2 \\ 0 & 0 & -1/2 \end{pmatrix}, \quad \det(zI - A) = (z - 1)(z + 1/2)^2. \quad (44)$$

The matrix is diagonalizable. Its 1-eigenspace is generated by e_0 , and its $-1/2$ eigenspace is

$$\left\{ ae_{-1} + \frac{a+c}{3}e_0 + ce_1 : a, c \in \mathbb{C} \right\}. \quad (45)$$

Taking real and imaginary combinations gives the basis in (7) after scaling from $\mathcal{P}$ to $\mathcal{L}$.

Proof of theorem 1.1 on the circle. The essential spectrum is theorem 2.1, scaled by one half. Outside that disk, every spectral value is an eigenvalue and every generalized eigenvector lies in F_1 by theorem 6.1. The finite matrix (44) therefore supplies all exterior values and their Jordan structure. The values 1 and $-1/2$ of $\mathcal{P}$ become $1/2$ and $-1/4$ for $\mathcal{L}$. The interior infinite-multiplicity statement scales in the same way. $\square$

7 Endpoint jets and interval spaces

The interval problem contains finite-dimensional information absent on the circle: endpoint derivatives need not match. For the whole Markov family, that information is exactly computable.

7.1 The endpoint quotient

For $f \in C^r([0, 1])$, define the mismatch functionals

$$\Delta_j(f) = f^{(j)}(1) - f^{(j)}(0), \quad 0 \leq j \leq r. \quad (46)$$

Their common kernel is the closed periodic-jet subspace

$$X_r^{\text{per}} = \{f \in C^r([0, 1]) : \Delta_j(f) = 0 \text{ for } 0 \leq j \leq r\}, \quad (47)$$

which is naturally isomorphic to $C^r(\mathbb{T})$ and has codimension $r + 1$. It is invariant under $\mathcal{P}_\omega$.

The quotient action follows from Leibniz' rule and the compatibility $\omega^{(k)}(1) = -\omega^{(k)}(0)$ for $k \geq 1$.

Lemma 7.1 (Exact endpoint-jet recursion). *For $0 \leq j \leq r$,*

$$\Delta_j(\mathcal{P}_\omega f) = 2^{-j}\omega(0)\Delta_j(f) + \sum_{k=1}^j \binom{j}{k} 2^{-(j-k)}\omega^{(k)}(0)\Delta_{j-k}(f). \quad (48)$$

Consequently the induced $(r + 1)$ -dimensional quotient matrix is triangular with diagonal

$$\omega(0), \frac{\omega(0)}{2}, \dots, \frac{\omega(0)}{2^r}. \quad (49)$$

Proof. Differentiate (12) j times. At derivative order $k = 0$, the midpoint terms agree at $x = 0$ and $x = 1$, while the endpoint terms leave $2^{-j}\omega(0)\Delta_j(f)$. For $k \geq 1$, the first-branch derivative at $x = 1$ is the negative of that at $x = 0$, and the derivative of the second weight has the opposite sign. The two midpoint contributions cancel, leaving exactly the k th summand in (48). $\square$

Theorem 7.2 (Endpoint-jet ladder). *On $C^r([0, 1])$,*

$$\sigma_{\text{ess}}(\mathcal{P}_\omega) = \overline{D}(0, 2^{-r}), \quad (50)$$

$$\sigma(\mathcal{P}_\omega; C^r([0, 1])) = \sigma(\mathcal{P}_\omega; C^r(\mathbb{T})) \cup \{\omega(0)2^{-j} : 0 \leq j \leq r\}. \quad (51)$$

Within the finite quotient, the endpoint values carry the algebraic multiplicities and internal collisions determined by the triangular jet matrix (48). A collision with a periodic resonance may also create a cross-block Jordan chain; the theorem asserts the complete spectral set, not a universal formula for that coupled Jordan structure.

Proof. Relative to a bounded complement of the invariant finite-codimensional subspace X_r^{per} , the interval operator is block upper triangular. Its restriction A is the circle operator, and its quotient C is the finite matrix of theorem 7.1. If both $A - \lambda I$ and $C - \lambda I$ are invertible, the usual triangular inverse formula shows that the full operator is invertible. This proves the inclusion of the interval spectrum in the displayed union.

For the reverse inclusion, first let $\lambda \in \sigma(A)$. If $|\lambda| \leq 2^{-r}$, then λ belongs to the interval essential spectrum because a finite-dimensional extension does not change that set. If $|\lambda| > 2^{-r}$, then $A - \lambda I$ is Fredholm of index zero; when it is noninvertible it has a nonzero kernel. The same vector, embedded in the periodic subspace, lies in the kernel of the full operator. Next let $\lambda \in \sigma(C)$. If $A - \lambda I$ is already noninvertible, the preceding argument applies. If it is invertible, choose $0 \neq y \in \ker(C - \lambda I)$ and solve $(A - \lambda I)x = -By$ for the off-diagonal block B ; then (x, y) is a nonzero kernel vector of the full block operator. Hence every value in both diagonal spectra survives. Essential spectrum is unchanged by the finite quotient, proving (50) as well. $\square$

Corollary 7.3 (Endpoint collapse for the sine mask). *For (13), $\omega_R(0) = 0$, so the quotient jet matrix is strictly triangular and nilpotent. Hence the interval and circle spectra of $\mathcal{P}$ coincide. After division by two, this completes theorem 1.1 on $C^r([0, 1])$.*

The formula (51) gives a precise answer to the repository's question about unmatched endpoint jets: they create a geometric ladder, and in the Rvachev case the endpoint zero collapses the entire ladder to the already present spectral value zero.

8 Optimal resonance expansions

8.1 Isolation thresholds

For $\mathcal{P}$, the essential radius on C^r is 2^{-r} . Thus the eigenvalue 1 is exterior exactly for $r \geq 1$, and $-1/2$ is exterior exactly for $r \geq 2$. Scaling by one half gives the thresholds in theorem 1.1. Since the exterior generalized eigenspaces lie in F_1 and the matrix (44) is semisimple, the Riesz projection ranks are one and two.

8.2 Proof of the sharp expansion

Proof of theorem 1.2. For $r \geq 2$, isolate the two exterior spectral points by disjoint small contours. The corresponding Riesz projections commute with $\mathcal{L}$ and have the ranks read from (44). On the complementary invariant subspace, the spectrum is exactly $\overline{D}(0, 2^{-r-1})$. The spectral radius formula gives (9) for every $\varepsilon > 0$. The lower bound is theorem 4.3, scaled by 2^{-n} ; subtracting the finite-rank resonance terms is a compact correction. $\square$

Remark 8.1 (No operator-norm superconvergence). The repository obtains scalar asymptotic refinements from special observables whose Fourier support or dual distribution has extra cancellation. The lower bound (10) shows that such improvement cannot hold uniformly over the full C^r unit ball after any finite-rank correction. A scalar observable may annihilate difficult directions; an operator norm cannot.

9 Comparison with analytic and Sobolev regimes

The same algebraic resonances appear in three distinct regularity geometries, but the noncompact component changes dramatically.

Space	Essential spectral set for $\mathcal{L}$	Complete spectral set	Mechanism
Analytic Hardy disk spaces	none away from 0; compact/trace-class regime	$\{0, 1/2, -1/4\}$	analytic compactness and finite Fourier determinant
Periodic $H^{s,\beta}$	$\overline{D}(0, 2^{-s}\sqrt{1+\sqrt{17}}/4)$	that disk $\cup \{1/2, -1/4\}$	square-mask coisometry and weighted Fourier geometry
Periodic or interval C^r	$\overline{D}(0, 2^{-r-1})$	that disk $\cup \{1/2, -1/4\}$	branch defect, derivative quotient, finite Fourier collapse

The Sobolev radius contains the factor $\sqrt{1+\sqrt{17}}/4$, which comes from the positive eigenfunction of a squared-mask transfer operator. The C^r radius does not. In a supremum-jet norm, the leading noncompact action is simply the highest derivative composed with inverse branches, producing 2^{-r} before the repository's outer factor $1/2$. The two formulas therefore encode different norm geometries, not conflicting calculations.

At the integer thresholds the contrast is concrete. In the Sobolev scale, the leading and subleading isolation thresholds are approximately 0.1785 and 1.1785. In the C^r scale they occur at the exact integers $r = 1$ and $r = 2$. In particular, $-1/4$ is on the essential boundary of C^1 , while it is strictly isolated on C^2 .

10 Verification, reproducibility, and formalization route

10.1 Exact finite checks

The accompanying standard-library Python program checks the algebraic parts of the argument without floating-point eigensolvers:

- the rational three-mode matrix, its characteristic polynomial, and the displayed eigenvectors;
- the descendant-frequency bound (39) over a finite stress range;
- the triangular endpoint-jet matrices through order twelve, including their exact diagonal ladders and sine-mask nilpotence;
- the scaling of the spectral thresholds and compact lower bounds listed in the summary table.

These checks are supplementary. No finite computation proves the infinite-dimensional Calkin theorem, the smoothness bootstrap, or the Fredholm alternative.

10.2 A staged Lean plan

A practical formalization can be divided into layers.

Layer 1: branch algebra. Define $\mathcal{P}_\omega$, the doubling operator $\mathcal{U}$, and prove $\mathcal{P}_\omega \mathcal{U} = \text{I}$. Formalize the explicit defect vector with compactly supported smooth input, or first prove the construction for a fixed smooth bump. The existing Fabius/Rvachev files already contain related transfer eigen-identities.

Layer 2: finite Fourier algebra. Formalize (37), invariance of F_1 , the exact matrix (44), and its eigenspaces. This layer is elementary and can be completed without a general spectral library.

Layer 3: endpoint jets. Define Δ_j and prove (48) by induction on the Leibniz formula. Finite-dimensional triangular-spectrum consequences then give the endpoint ladder.

Layer 4: Banach and Fredholm infrastructure. Formalize the periodic derivative quotient, compactness of $C^r \hookrightarrow C^{r-1}$, invariance of the Fredholm essential spectrum under compact perturbations, and the Calkin spectral mapping theorem. This is the largest reusable investment. Once available, the universal disk statement becomes short.

Layer 5: exterior regularity and Fourier collapse. Formalize power boundedness, the resolvent bootstrap, rapid Fourier decay of smooth periodic functions, and the descendant-frequency estimate. This connects the general essential theorem to the complete special spectrum.

No Lean source is included in the present package, and no repository build was run. The plan distinguishes elementary algebraic targets from the substantial functional-analytic library work.

11 Further research questions and topics

Question 11.1 (Noninteger Hölder and Zygmund scales). Determine the complete spectrum on $C^{r,\alpha}(\mathbb{T})$ for $0 < \alpha < 1$, on little-Hölder spaces, and at the Zygmund endpoints. The defect eigenseries predicts an essential radius at least $2^{-(r+\alpha+1)}$ for $\mathcal{L}$, and standard Lasota–Yorke estimates suggest the matching upper radius. The difficult part is a complete exclusion or classification of finite-regularity eigenfunctions in the annulus between successive integer thresholds.

Question 11.2 (Boundary point spectrum at C^1). Classify the eigenspace and approximate eigenspaces of $-1/4$ on C^1 . The value is simultaneously a smooth eigenvalue and a point of the essential boundary. Is its geometric multiplicity finite or infinite? What is the local resolvent growth, and which defect vectors generate boundary approximate eigenfunctions?

Question 11.3 (General trigonometric Markov masks). For smooth trigonometric ω satisfying $\omega(x+1) = 1 - \omega(x)$, classify all spectral values outside $\overline{D}(0, 2^{-r})$ by a finite Fourier matrix. Determine a sharp band size in terms of the degree and dilation, and characterize masks for which all exterior eigenvalues are semisimple.

Question 11.4 (Endpoint-ladder collisions). In (51), analyze what happens when an endpoint value $\omega(0)2^{-j}$ collides with a periodic resonance. Give necessary and sufficient conditions for a Jordan chain crossing the periodic and jet blocks, and derive perturbation expansions through such a collision.

Question 11.5 (Weighted endpoint spaces). Replace the ordinary C^r norm by weights that penalize or permit specified endpoint singularities. Which portions of the jet ladder become essential, and when does the finite quotient turn into an infinite boundary spectrum? A satisfactory theorem should include power and logarithmic weights.

Question 11.6 (Exact Riesz projectors). Compute the leading and alternating Riesz projectors of the Rvachev operator as explicit functionals on C^r , rather than contour integrals. The ranges are known from (7); the open problem is an exact, well-conditioned description of the dual coefficient functionals and their relation to the Thue–Morse and Stern distributions in the repository.

Question 11.7 (Sharp resolvent and pseudospectral bounds). Determine the asymptotic size of $\|(\lambda I - \mathcal{L})^{-1}\|$ as λ approaches the essential circle, especially near $-1/4$ in C^1 . The explicit right inverse and defect series may permit two-sided bounds and reveal whether the pseudospectrum is approximately circular or strongly distorted by the finite resonances.

Question 11.8 (Nonlinear expanding maps). Replace doubling by a smooth nonlinear expanding degree-two map. The highest derivative then carries a variable factor $|T'|^{-r}$. Find hypotheses under which the full Fredholm essential spectrum, not merely its radius, is an explicit annulus or disk determined by a weighted composition cocycle.

Question 11.9 (Higher-dimensional toral maps). For an expanding integer matrix on a torus, construct the analogue of the branch-cancellation defect and determine the C^r essential spectral body. On a cube, endpoint jets should be replaced by a hierarchy of face, edge, and corner jets. Identify the resulting finite spectral ladders and their collision rules.

Question 11.10 (Optimal scalar asymptotics at the operator boundary). The repository’s Rvachev moments exhibit a scalar alternating correction even when the corresponding eigenvalue lies on an essential boundary. Classify the continuous linear functionals on C^1 that annihilate enough defect directions to admit an $o(4^{-n})$ remainder. Determine whether the known Fourier cutoff is necessary or merely one sufficient mechanism.

Question 11.11 (Proof-producing spectral certificates). Build a verified interface that accepts a rational trigonometric Markov mask, constructs its finite Fourier block and endpoint-jet matrix, and returns a certificate for all exterior resonances once the universal Fredholm theorem has been formalized. The output should distinguish isolated eigenvalues, boundary eigenvalues, and collisions with the endpoint ladder.

12 Conclusion

The finite-smoothness question in the Rvachev spectral branch has an exact integer-order answer. On C^r , analytic spectral collapse is replaced by a closed disk of radius 2^{-r-1} , together with the same two algebraic resonances $1/2$ and $-1/4$. The disk is not an artifact of an estimate: its interior consists of infinite-multiplicity eigenvalues, its boundary is forced by closedness of the Fredholm essential spectrum, and its radius gives a sharp lower bound against every compact approximation.

The proof reveals a reusable structure. A dyadic Markov partition gives a right inverse; two inverse branches give an infinite defect space; and the highest derivative conjugates the Calkin image by the exact scale 2^{-r} . For interval spaces, every unmatched endpoint jet is visible in an explicit triangular quotient, producing a geometric ladder. The Rvachev sine mask vanishes at the endpoint, so that ladder collapses and the circle and interval spectra agree.

This synthesis links four levels of information that had previously appeared separately in the repository: branch identities, finite Fourier algebra, finite-smoothness Fredholm geometry, and

resonance asymptotics. It also isolates the next genuinely difficult frontier: noninteger Hölder and endpoint-weighted spaces, where the essential radius is plausible but the complete exterior and boundary spectral structure is not yet settled here.

A Fredholm facts used in the proof

For completeness, the functional-analytic inputs are listed precisely.

1. The set of Fredholm operators between Banach spaces is open, and the Fredholm index is locally constant.
2. Compact perturbations preserve Fredholmness and index; equivalently, the Fredholm essential spectrum is the spectrum of the Calkin image.
3. Similar operators have the same spectrum and essential spectrum.
4. A finite-dimensional invariant extension does not alter Fredholm essential spectrum. A block triangular operator is invertible whenever both diagonal blocks are invertible. Equality with the union of the diagonal spectra requires an additional argument; the finite-quotient and index-zero conditions needed here are supplied in the proof of [theorem 7.2](#).
5. The spectrum of an element in a unital Banach algebra obeys spectral mapping, and its norm dominates its spectral radius.
6. Outside the Fredholm essential spectrum, on a connected component containing infinity, a noninvertible operator of index zero has a nonzero finite-dimensional kernel. Isolated spectral values have finite-rank Riesz projections.
7. The embedding $C^m(\mathbb{T}) \hookrightarrow C^{m-1}(\mathbb{T})$ is compact for $m \geq 1$ by Arzela–Ascoli.

All operator-specific identities needed to invoke these facts are proved in the body of the article. Standard references include Kato [8], Nussbaum [7], and Hennion [6].

B Verification program and recorded scope

The file `verify.py` uses only the Python standard library. Its recorded run writes `verification.json`. The default finite ranges are deliberately modest and exact. The program checks:

Check family	Recorded count or range
Three-mode matrix identities	all 9 entries
Characteristic polynomial coefficients	all coefficients
Displayed eigenvectors	3 exact vectors
Frequency descendants	levels 0 through 12
Jet matrices	orders 0 through 12
Nilpotence for the sine endpoint	orders 0 through 12
Threshold table	orders 0 through 12

The finite program does not sample the infinite-dimensional spectrum. Its role is to catch transcription, sign, indexing, and scaling errors in the parts of the proof that reduce to exact finite algebra.

C Repository provenance and claim boundaries

The comparison target is the public repository

<https://github.com/VladimirReshetnikov/ProveIt>

at inspected main-branch commit

ffddaa8b9c89e7bf027e1442cc6216bb010906d0.

The relevant current paths and source identities were:

Repository item	Role in this article
.../rvachev_up_fourier_decay/ README.md 6f31c8b9c60d8a2fc10e b8fb55e8a752672efe42	Group-level status, analytic and Sobolev arrivals, and explicit note that Hölder spaces remain untreated.
.../Spectral_Collapse_ Alternating_RMS/article.tex cad8f114f2298bd46907c 451f84ace56f93fa784	Source of the question “Spectra at finite smoothness” and the analytic three-point spectrum.
.../Sobolev_Spectral_Disks_ Rvachev_Thue_Morse/article.tex ff81881e993a5400489715 8c8e6a11da5ebc1b69	Exact Sobolev disk and explicit statement that Hölder/ C^r remains open.
Analysis/FabiusFunction/Lean/ FabiusFunction/	Existing machine-checked transfer eigen-identities and related Fabius infrastructure; no new build or theorem is claimed here.

The ellipses abbreviate the common path

Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/.

The repository was changing rapidly on 30 September 2026. The immutable commit and blob identifiers prevent later editorial movement from being mistaken for the exact comparison corpus.

The literature check located directly relevant work on transfer operators on the circle with trigonometric sine and cosine weights [3], general weighted-isometry disk phenomena [4], and the classical smooth/Hölder transfer-operator framework [5, 6]. This is not an exhaustive worldwide priority search. Accordingly, the manuscript claims a rigorous resolution of the stated repository gap and a self-contained structural theorem, not certified global first publication.

References

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